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CSCI900 Quiz 5

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~~1~~ Give an example of a symmetric 4x4 matrix.

Let $A = \begin{bmatrix} 2 & -3 \\ 1 & -2 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 \\ 1 & 3 \end{bmatrix}$, $C = \begin{bmatrix} 3 & 0 & 1 \\ -1 & 2 & 4 \end{bmatrix}$

2. Find $A+B$

$$\begin{bmatrix} 3 & -1 \\ 2 & 1 \end{bmatrix}$$

3. Find AB

$$\begin{bmatrix} -1 & -5 \\ -1 & -4 \end{bmatrix}$$

4. Find BA

$$\begin{bmatrix} 4 & -7 \\ 5 & -9 \end{bmatrix}$$

5. Find CB $2 \times 3 \times 2 \times 2$

not possible

6. Find BC $2 \times 2 \times 2 \times 3$

$$\begin{bmatrix} 1 & 4 & 9 \\ 0 & 6 & 13 \end{bmatrix}$$

7. Find C^T

$$\begin{bmatrix} 3 & -1 \\ 0 & 2 \\ 1 & 4 \end{bmatrix}$$

8. Find $(AB)^T$

$$\begin{bmatrix} -1 & -1 \\ -5 & -4 \end{bmatrix}$$

9. Find $B^T A^T$

$$\begin{bmatrix} -1 & -1 \\ -5 & -4 \end{bmatrix}$$

10. Find A^{-1} and B^{-1}

$D = ad - bc$
 $D = -4 - (-3)$
 $D = -1$

$$A^{-1} = \begin{bmatrix} 2 & -3 \\ 1 & -2 \end{bmatrix}$$

$D = 3 - 2$
 $D = 1$

$$B^{-1} = \begin{bmatrix} 3 & -2 \\ -1 & 1 \end{bmatrix}$$